

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – APPLICATION BASED QUESTIONS (5 Questions)

Course Code:	DSA0605- SLOT - C	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Precision (BL3)	Assessment:	Assessment Tool 2 – Application Based Questions
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO1 Statement:	Apply appropriate visualization methods to represent distributions, proportions, and relationships in data (BL3).		
Student Name:	G Kushwanth	Reg. No.:	192424150
Section / Batch:	2024-AI&DS	Date:	

Code of Conduct

I **G Kushwanth** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: G Kushwanth

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Amounts & Proportions	Category-wise comparison and proportion analysis using bar charts	1	5
Distributions	Analyzing data spread and patterns using histograms, density plots, boxplots	2	5

Temporal & x-y Relationships	Studying trends over time and relationships between variables	3	5
Geospatial & Temporal Data	Mapping locations and analyzing delivery patterns over time	4	5
Multivariate Data Visualization	Exploring multiple variables and comparing distributions	5	5
GRAND TOTAL:		25 Marks	

Annexure A – Application Based Questions

1.A university wants to analyze student enrollments across various departments such as Computer Science, Mechanical Engineering, Civil Engineering, and Electronics. The administration aims to identify departments with the highest enrollments, compare enrollment trends over multiple semesters, and understand the proportion of students in each department. How would you categorize the data and choose appropriate visualizations to represent enrollment amounts, proportions, and temporal trends across departments?

Answer

The university's enrollment data contains three types of information:

- **Categorical Data:** Department names (Computer Science, Mechanical, Civil, Electronics).
- **Numerical Data:** Number of students enrolled.
- **Temporal Data:** Semester-wise enrollment records.

To analyze this data effectively:

1. Bar Chart

- Used to compare the total number of students enrolled in each department.
- Helps identify the department with the highest and lowest enrollment.

2. Pie Chart

- Represents the proportion of students in each department.
- Shows the percentage contribution of every department to the total enrollment.

3. Line Chart

- Displays semester-wise enrollment trends.
- Helps determine whether admissions are increasing, decreasing, or remaining stable over time.

Conclusion

Using these visualizations enables the university to compare departmental enrollments, monitor enrollment growth across semesters, and understand the distribution of students. The insights support better admission planning, faculty allocation, and resource management.

2.A manufacturing company records the number of defects found in products produced by different production lines. The quality assurance team wants to examine the distribution of defects, identify variability among production units, and detect unusual patterns. How would you use histograms, density plots, violin plots, and boxplots to analyze defect distributions and compare quality performance across production lines?

Answer

The manufacturing company records defects from multiple production lines to evaluate product quality.

The dataset consists of:

- **Categorical Data:** Production line names.
- **Numerical Data:** Number of product defects.

The following visualizations are suitable:

1. Histogram

- Displays the frequency distribution of defects.
- Reveals whether defects are concentrated within a specific range.

2. Density Plot

- Shows the overall distribution pattern.
- Helps determine whether the defect data is symmetric, skewed, or multimodal.

3. Violin Plot

- Compares defect distributions among production lines.
- Displays both density and spread of the data.

4. Boxplot

- Identifies median, quartiles, variability, and outliers.
- Helps detect production lines with unusually high defect counts.

Conclusion

These visualizations help quality engineers identify inconsistent production lines, detect abnormal defect patterns, compare production quality, and implement corrective actions to improve manufacturing efficiency.

3.An online learning platform tracks student activity including time spent on courses, number of videos watched, quiz scores, and course completion rates. The platform seeks to understand relationships among these variables and identify factors influencing successful course completion. How would you use scatter plots, heatmaps, and multivariate visualizations to analyze relationships among learner engagement metrics and academic outcomes?

Answer

The online learning platform collects learner engagement metrics such as:

- Time spent on courses
- Number of videos watched
- Quiz scores
- Course completion rates

The data includes numerical variables that can be analyzed for relationships. The following visualizations are appropriate:

1. Scatter Plot

- Shows the relationship between study time and quiz scores.
- Identifies whether increased study time improves academic performance.

2. Heatmap

- Displays correlations among variables such as study time, videos watched, quiz scores, and completion rates.
- Helps identify the strongest positive or negative relationships.

3. Multivariate Visualization

- Simultaneously compares multiple learner metrics.
- Reveals how different engagement factors collectively influence course completion.

Conclusion

These visualizations enable educators to identify important learning behaviors, predict student success, improve course design, and increase completion rates through data-driven interventions.

4.A city administration collects traffic data including vehicle counts, congestion levels, accident locations, and travel times across different regions and time periods. The objective is to identify traffic hotspots and improve transportation planning. How would you represent geospatial and temporal data using maps, heatmaps, and time-series visualizations to identify traffic patterns and optimize road management?

Answer

The city administration collects traffic information including:

- Vehicle counts
- Congestion levels
- Accident locations
- Travel times
- Geographic regions
- Time periods

The following visualizations effectively represent the data:

1. Geographical Map

- Displays accident locations and traffic distribution across city regions.
- Identifies high-risk and high-traffic zones.

2. Traffic Heatmap

- Uses color intensity to represent congestion levels.
- Quickly highlights traffic hotspots requiring attention.

3. Time-Series Line Chart

- Shows traffic volume over different time periods.
- Helps identify peak traffic hours and daily traffic patterns.

Conclusion

These visualizations help city planners identify congestion hotspots, optimize traffic signal timing, improve road infrastructure, reduce travel time, and enhance transportation planning.

5.A large organization records communication data among employees through emails, meetings, and project collaborations. Management wants to identify key employees, collaboration groups, and communication bottlenecks within the organization. How would you use network visualizations and multivariate analysis techniques to represent communication patterns, identify influential individuals, and improve organizational collaboration?

Answer

The organization records communication through:

- Emails
- Meetings
- Project collaborations

The objective is to analyze communication patterns and collaboration efficiency. The following visualization techniques are suitable:

1. Network Visualization

- Represents employees as nodes and communication links as edges.
- Identifies highly connected employees, collaboration groups, and communication bottlenecks.

2. Multivariate Analysis

- Combines communication frequency, meeting participation, and project involvement.
- Enables comparison of multiple employee performance indicators simultaneously. **Conclusion**

Network visualization and multivariate analysis help management identify influential employees, strengthen collaboration between teams, reduce communication bottlenecks, and improve organizational productivity through informed decision-making.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem		
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts		
Problem Solving Approach	20%	Logical, well structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach		
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution		
Clarity	10%	Clear, well organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____